

Java Programming Notes - Questions and Answers

This PDF contains important Java programming concepts in Question and Answer format. These notes are useful for exams, viva, interviews, and academic preparation.

Q: 1. What is Java?

Answer: Java is a high-level, object-oriented programming language developed by Sun Microsystems. It is platform independent because Java programs run on the Java Virtual Machine (JVM). Java follows the principle of Write Once, Run Anywhere (WORA).

Q: 2. What are the features of Java?

Answer: The main features of Java are simplicity, platform independence, security, robustness, multithreading, portability, and automatic garbage collection.

Q: 3. What is JVM?

Answer: JVM stands for Java Virtual Machine. It is responsible for executing Java bytecode. JVM makes Java platform independent.

Q: 4. What is the difference between JDK, JRE, and JVM?

Answer: JDK is used for development and includes compiler tools. JRE provides the runtime environment. JVM executes Java bytecode.

Q: 5. What are primitive data types in Java?

Answer: Java primitive data types are int, float, double, char, boolean, byte, short, and long.

Q: 6. What is a variable in Java?

Answer: A variable is a container used to store data values during program execution.

Q: 7. What are operators in Java?

Answer: Operators perform operations on variables and values. Types include arithmetic, relational, logical, assignment, and unary operators.

Q: 8. Explain if-else statement.

Answer: The if-else statement is used for decision making. It executes different blocks of code based on conditions.

Q: 9. What is a loop?

Answer: A loop repeatedly executes a block of code until a condition becomes false. Java supports for, while, and do-while loops.

Q: 10. What is an array?

Answer: An array is a collection of elements of the same data type stored in contiguous memory locations.

Q: 11. What is a String in Java?

Answer: A String is a sequence of characters. Strings in Java are immutable objects.

Q: 12. Difference between String and StringBuilder.

Answer: String objects are immutable while StringBuilder objects are mutable and faster for modifications.

Q: 13. What is Object-Oriented Programming?

Answer: Object-Oriented Programming (OOP) is a programming paradigm based on objects and classes. Main concepts are inheritance, polymorphism, abstraction, and encapsulation.

Q: 14. What is a class?

Answer: A class is a blueprint used to create objects. It contains variables and methods.

Q: 15. What is an object?

Answer: An object is an instance of a class that contains properties and behaviors.

Q: 16. What is a constructor?

Answer: A constructor is a special method used to initialize objects. It is automatically called when an object is created.

Q: 17. What is inheritance?

Answer: Inheritance allows one class to acquire properties and methods of another class. It improves code reusability.

Q: 18. What is polymorphism?

Answer: Polymorphism allows one method or object to take multiple forms. It includes method overloading and method overriding.

Q: 19. What is abstraction?

Answer: Abstraction hides implementation details and shows only essential features. It is achieved using abstract classes and interfaces.

Q: 20. What is encapsulation?

Answer: Encapsulation binds data and methods together into a single unit. It provides data security through access modifiers.

Q: 21. What is an interface?

Answer: An interface contains abstract methods that must be implemented by classes. Java supports multiple inheritance through interfaces.

Q: 22. What is exception handling?

Answer: Exception handling is used to manage runtime errors. Java uses try, catch, throw, throws, and finally blocks.

Q: 23. What is multithreading?

Answer: Multithreading allows multiple threads to execute simultaneously. It improves application performance.

Q: 24. What is the Java Collections Framework?

Answer: The Collections Framework provides classes and interfaces for storing and manipulating data. Examples include ArrayList, HashSet, and HashMap.

Q: 25. What is JDBC?

Answer: JDBC stands for Java Database Connectivity. It is used to connect Java applications with databases.

Q: 26. What is garbage collection?

Answer: Garbage Collection automatically removes unused objects from memory to free resources.

Q: 27. What are access modifiers in Java?

Answer: Access modifiers control the visibility of classes, methods, and variables. Types are public, private, protected, and default.